

Glass



What is glass?

An amorphous fusion of mineral compounds that produces a transparent solid when cooled. A 3D network of atoms that lacks the repeated, orderly arrangement typical of crystalline materials.

Physical properties:

- hard, brittle, transparent, translucent
- insulative (thermal and electrical)
- supercooled liquid of infinite viscosity,
- can absorb, transmit and reflect light

Chemical: resistant to all but fluorine and very strong bases.

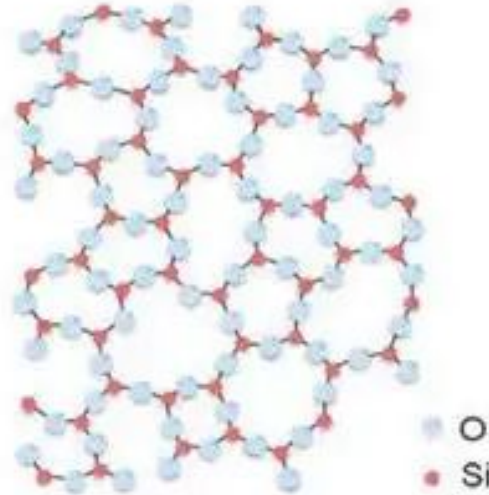
- **Transparency:** Most glass allows light to pass through, making it ideal for windows, lenses, and decorative uses.
- **Translucency:** Translucent materials allow some light to pass through, but not all. Because of this, objects seen through them appear blurred or unclear.
- **Hardness:** Glass is relatively hard (around 5.5–7 on the Mohs scale), but it is also brittle, meaning it fractures easily under stress.
- **Brittleness:** Unlike metals, glass does not deform plastically; it breaks suddenly when stressed.
- **Insulator:** Glass is an excellent electrical insulator, widely used in electronics and power systems.
- **Chemical Resistance:** Glass resists most acids but can be attacked by hydrofluoric acid.
- **Non-reactive:** It doesn't corrode or rust, making it ideal for containers and lab equipment.

Amorphous structure of glass

Glass is called an amorphous solid because its atoms are arranged in a random fashion.

Because of its irregular pattern, when glass is broken, it produces a variety of fracture patterns.

Mainly crystalline objects like glass or quartz



Glass evidence at crime scene

Automobile accident sites may be littered with broken headlight or windshield glass.

The site of a store break-in may contain shards of window glass with fibers or blood on them.

If shots are fired into a window, the sequence and direction of the bullets can often be determined by examining the glass. Minute particles of glass may be transferred to a suspect's shoes or clothing and can provide a source of trace evidence linking a suspect to a crime.

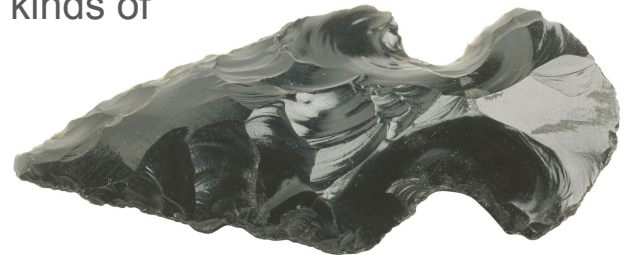
How was glass made ?

Long before humans began making glass, glass formed naturally. When certain types of rock are exposed to extremely high temperatures, such as lightning strikes or erupting volcanoes, glass can form.

Prehistoric humans used obsidian as a cutting tool. The earliest man-made glass objects (glass beads) were found in Egypt, dating back to 2500 BC.

Glass blowing began sometime during the first century BC. By the 14th century, knowledge of glassmaking had spread throughout Europe.

The Industrial Revolution brought the mass production of many kinds of glass.



How is glass formed?

Glass is a hard, brittle, amorphous material made by melting sand (aka silica, silicon dioxide, SiO_2), lime (aka calcium oxide, CaO), and soda (sodium carbonate, Na_2CO_3) at very high temperatures (1,400–1,600 °C).

The lime (CaO) is added to prevent the glass from being soluble in water.

Na_2CO_3 —soft glass—e.g., soda lime glass- $\text{Na}_2\text{O}.\text{CaO}.6\text{SiO}_2$

K_2CO_3 - hard glass- e.g., potash lime glass- $\text{K}_2\text{O}.\text{CaO}.6\text{SiO}_2$

Types of glass

1. Soda-lime glass
2. Float glass
3. Borosilicate glass
4. Leaded glass
5. Tempered glass
6. Laminated glass
7. Bulletproof glass

Soda-lime glass

Mostly sand, sodium carbonate, and calcium oxide.

Silica (SiO_2): ~60–75%

Soda (Na_2O): ~12–18%

Quicklime (CaO): ~12–15%

Colors are due to impurity.

Used in manufacturing most window bottle glass, glass tubes, bulbs,



Float/flat glass

Float glass is typically used for windows.

Soda-lime glass that has been cooled on top of a bath of molten tin. Mixed and melted at 1500°C.

It can resist pressure up to 10 kg. But it breaks from sudden impact.



Borosilicate glass

Cookware, laboratory glassware, light bulbs

Lime is replaced with boron oxide (B_2O_3).

Increases resistance to thermal shock.

E. g. Pyrex®



Leaded glass

Fine glassware and decorative art glass, called crystal or leaded glass, substitutes lead oxide (PbO) for calcium oxide (lime).

The addition of lead oxide makes the glass denser.

As light passes through the more dense glass, the light waves are bent, giving the glass a sparkling effect.

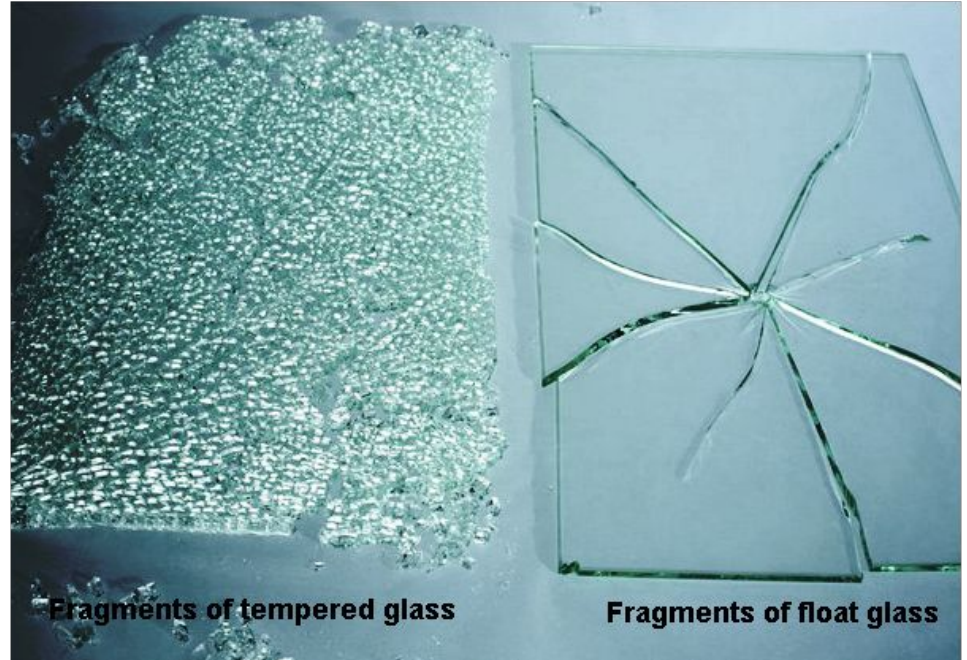


Tempered glass

Treated by heating and quickly cooling float glass OR chemically treating the surface, replacing the Na^+ with K^+

Designed to be much tougher and shatter into tiny pieces rather than fracture.

Used in auto side windows and high-stress areas.



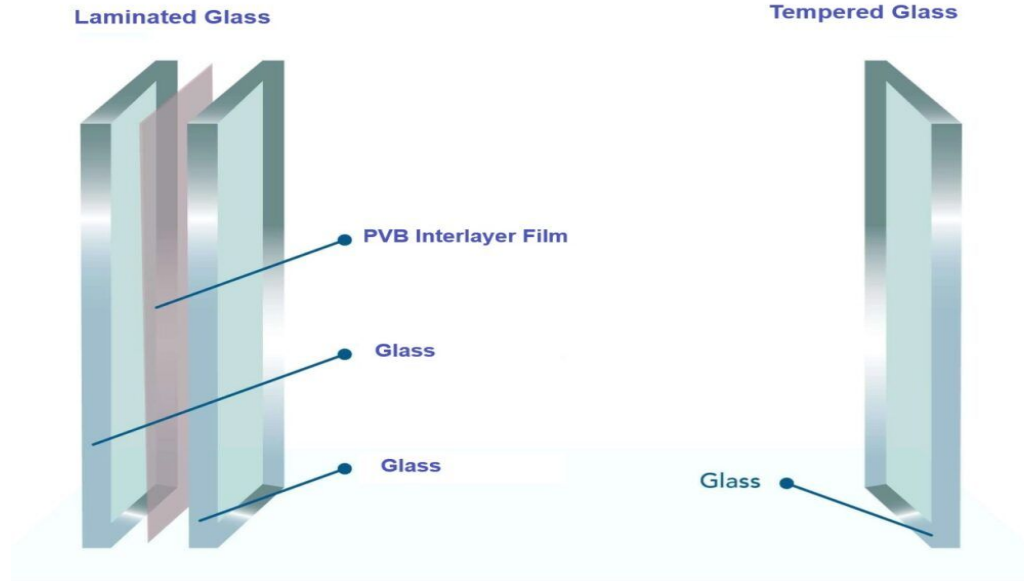
Laminated glass

Laminated glass is a type of safety glass made by bonding two or more layers of glass with a plastic interlayer (commonly Polyvinyl Butyral—PVB).

Holds fragments together after breakage, reducing injury risk.

Breaks with a “spider web” cracking pattern instead of sharp shards.

Used in windshields, skylights, storefronts, and protective glazing.



Bulletproof glass

Bulletproof glass is a combination of two or more types of glass, one hard and one soft. The softer layer makes the glass more elastic so it can flex instead of shatter.

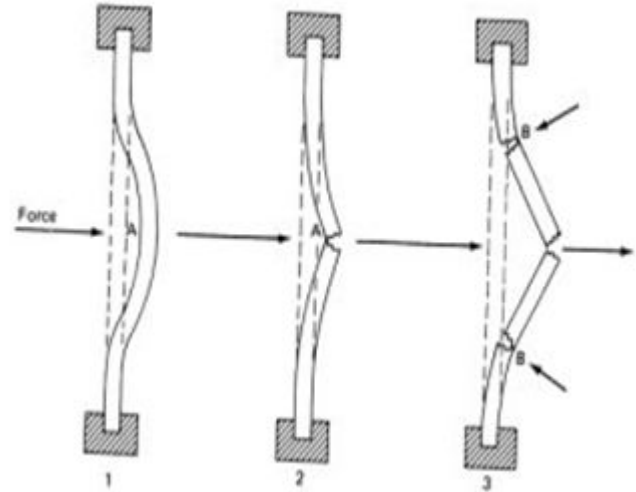
The index of refraction for both of the glasses used in the bulletproof layers must be almost the same to keep the glass transparent and allow a clear view through the glass. Bulletproof glass varies in thickness from three-quarters of an inch to three inches.



How does glass break?

Each force causes a deformation that may leave a visible mark or fracture on the glass. This can be used to determine the direction and amount of force.

Glass acts initially as an elastic surface and bends away when a force is applied. When the force increases beyond its tensile strength, it cracks.



Fracture in glass

Glass has some flexibility. When glass is hit, it can stretch slightly. When glass is forced to stretch too far, fracture lines appear and the glass may break.

The fracture patterns on broken glass can provide clues about the direction and rate of impact.

Fracture is caused by small imperfections, flaws, and irregularities on the surface of the glass. Flaws are very fine cracks that cause a concentration of stress, and the crack proceeds quickly, causing a fracture.

Primary Fractures: radial

When glass breaks, fracture patterns form on the surface.

The radial fractures start at the point of impact and radiate (like spokes on a wheel) outward from there.

Radial fractures form on the side opposite the point of impact.

tension = pull = opposite side of
impact=radial

Secondary Fractures: concentric

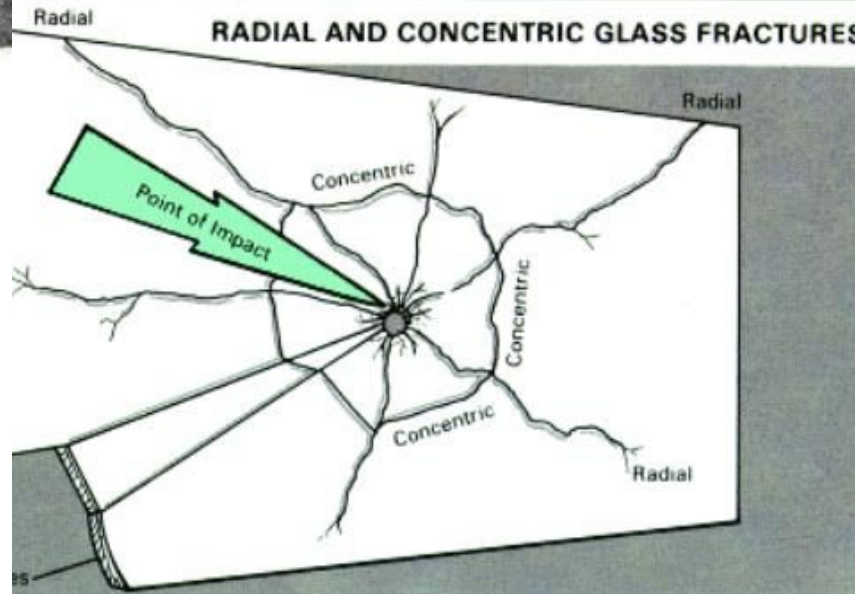
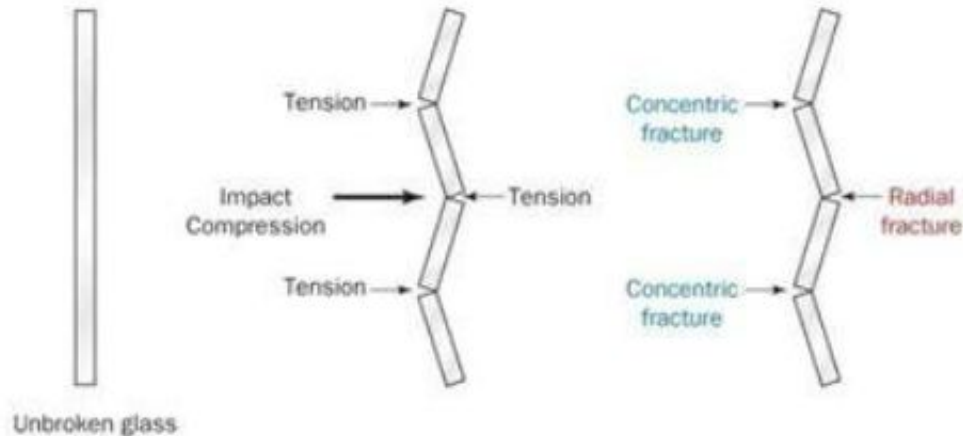
Secondary fractures may also form.

These fractures take the form of concentric circles around the point of impact.

Concentric circles are circles that have the same center. Concentric circles form on the same side of the glass as the point of impact.

Compression = push = same side of
impact = concentric

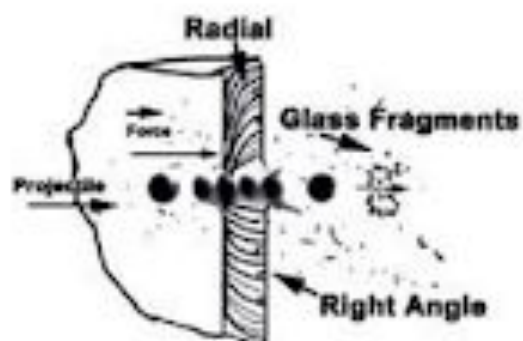
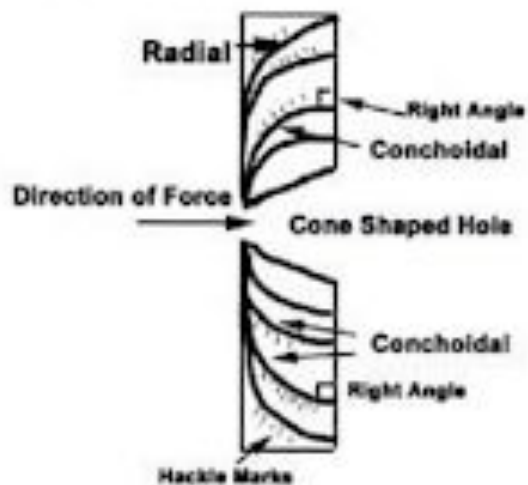
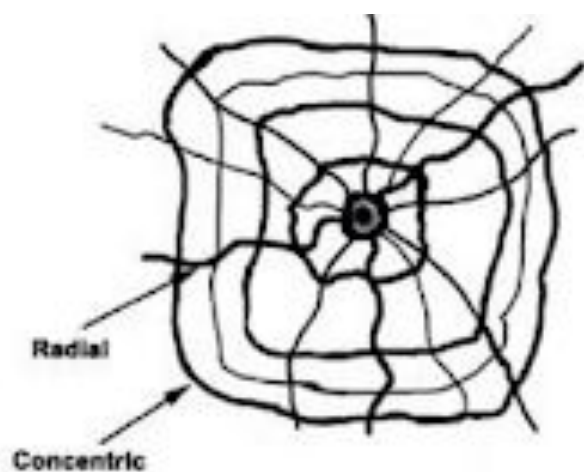
Why Radial and Concentric Fractures Form



Rib marks (stress lines) form right angles on the reverse side of the impact—this is known as the 3R Rule: Ridge lines on **Radial fractures** make **Right angles** to the **Rear**.

Exceptions

- tempered glass “dices” without forming ridges
- very small windows held tightly in the frame can’t bend or bulge appreciably
- windows broken by heat explosions have no “point of impact” has smooth curved edges at impact

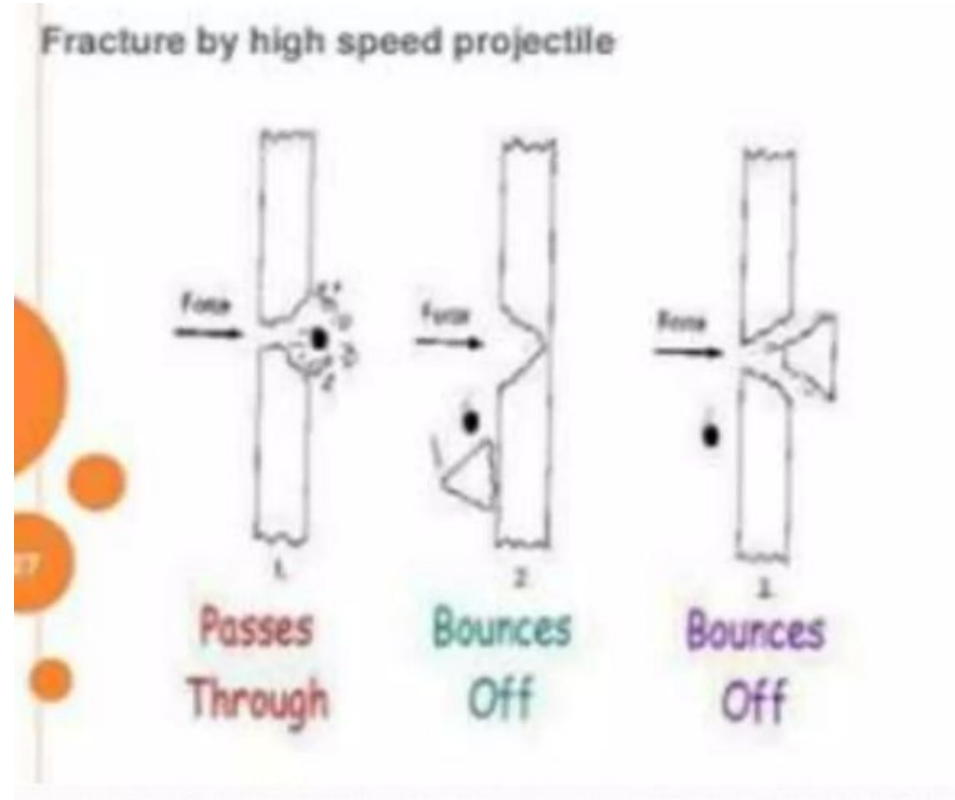


**Right - Rear - Radial
(Direction of Force)**

Cone or Crater Fractures

Funnel-shaped damage caused by high-velocity impacts (e.g., bullets).

The wider part of the cone is on the exit side, indicating the direction of travel



Rib Marks :

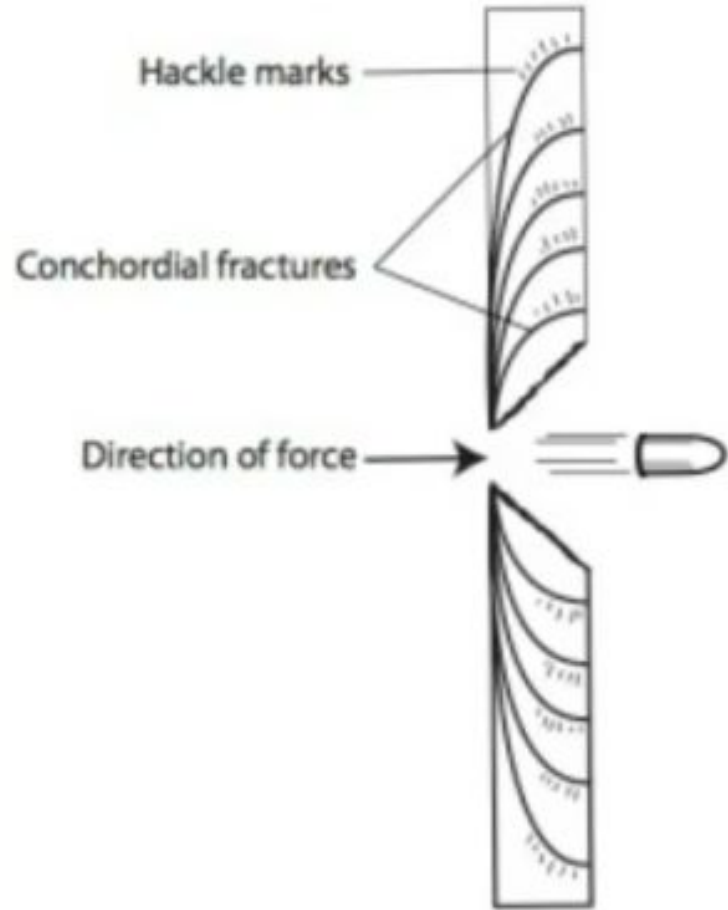
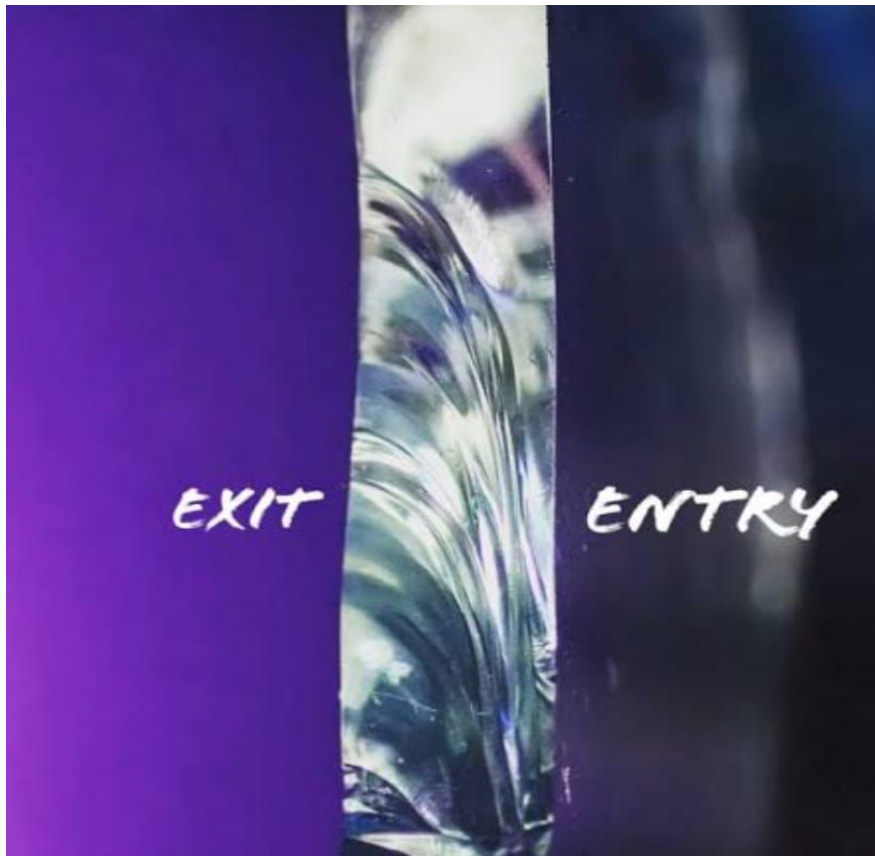
Rib marks help determine the direction of force.

The side with perpendicular rib marks is the side opposite the force.

Hackle Marks :

Formation: As a crack moves through glass, it leaves behind these marks in the direction of travel.

Appearance: Hackle marks radiate outward from the point of impact and curve slightly in the direction the crack was travelling.



Rib marks

Breakage of glass from fire

During a fire, glass may break as a result of heat fracturing.

Heat fracturing produces breakage patterns on glass that are different from breakage patterns caused by impact.

Wavy fracture lines develop in glass that has been exposed to high heat. (v-shaped)

Glass will tend to break toward the region of higher temperature. If the glass was not broken before the fire, then it will lack both radial and concentric fractures.

Edges may be smooth or curved.



Successive penetration on glass

When there have been successive penetrations of glass, it is frequently possible to determine the sequence of impact by observing the existing fracture lines and their points of termination.

A fracture always terminates at an existing line of fracture.

Pic: The hole on the right was formed first. When cracks radiating from the hole come into contact with another crack, they stop. Stress applied to the crack that is causing it to crack will be distributed along the existing crack rather than across it.



The **exit hole** of a projectile going through a glass is **always larger than** the **entry hole** because of the shock waves traveling out in a concentric pattern from the impact point.

Backward fragmentation refers to the phenomenon where glass fragments are ejected in the direction opposite to the incoming force—especially in high-velocity impacts like gunshots or explosions.

Mechanism: When a projectile strikes glass, it compresses the material at the point of impact. The energy causes the glass to fracture and eject fragments backward, toward the source of the force.

Forensic Significance: The entry side of the glass typically has a smaller hole with more backward fragmentation. The exit side shows a larger, more conical hole with fewer fragments. Investigators can use the distribution of glass particles at a scene to infer the direction of the projectile or explosion.

Also known as backscatter, which helps as trace evidence to catch the perpetrator.

Examination techniques

Callipers and micrometers—for measuring thickness

Microscopic Analysis—Stereomicroscopes and polarized light microscopes are used to observe fracture patterns and surface features.

Refractive Index (RI) Testing – Measures how light bends through the glass. Glass fragments are compared using the GRIM (Glass Refractive Index Measurement) system.

Elemental Analysis – X-ray fluorescence (XRF) identifies the elemental composition of glass. Helps match fragments to known sources.

UV Light Examination—Short- and long-wave UV light can reveal coatings or treatments on glass surfaces

Density

Each type of glass has a density that is specific to that glass. One method of matching glass fragments is by density comparison. To determine the density of glass, it is best to use the floatation method.

Density = mass/volume

Mass is measured on a balance/scale

Volume is determined by immersing parts of the glass and seeing the level change in a measuring glass.

Principle: A solid particle floats, sinks, or remains suspended depending on its density relative to the liquid.

Method: A reference glass chip is placed in a liquid mixture (bromoform + bromobenzene). The liquid's composition is adjusted until the chip stays suspended, meaning the liquid and glass have equal density. Unknown glass chips of similar size/shape are added for comparison. If they also remain suspended, their density matches the reference; otherwise, they sink or float.

Precision: Window glass density varies internally by up to 0.0003 g/mL. To avoid confusion between natural variation and different origins, sensitivity should not exceed this value. The flotation method can distinguish density differences as small as 0.001 g/mL, making it reliable for forensic comparison.

Density of solids: (glass) 2.4 - 2.8 g/cm³

Density of liquids: (water) 1 g/cm³

Lighter liquid—
Xylene, bromobenzene,
Benzene, nitrobenzene

Heavier liquid-
Bromoform
Methylene iodide

Limitations and Considerations:

Density measurements are generally performed when elemental analysis is not available.

It's impractical to measure density on very small glass fragments (less than 2 mm).

Use of hazardous liquid, i.e., bromoform.

Measurements can be inaccurate for small, irregular, or dirty fragments.

Density measurements are now rarely performed in forensic laboratories, largely replaced by refractive index measurements, which are faster, more accurate, and more precise.

Refractive index

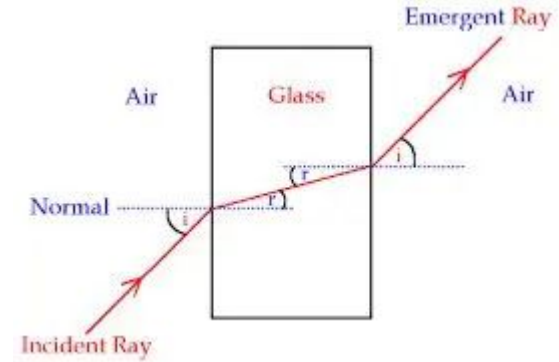
Refraction

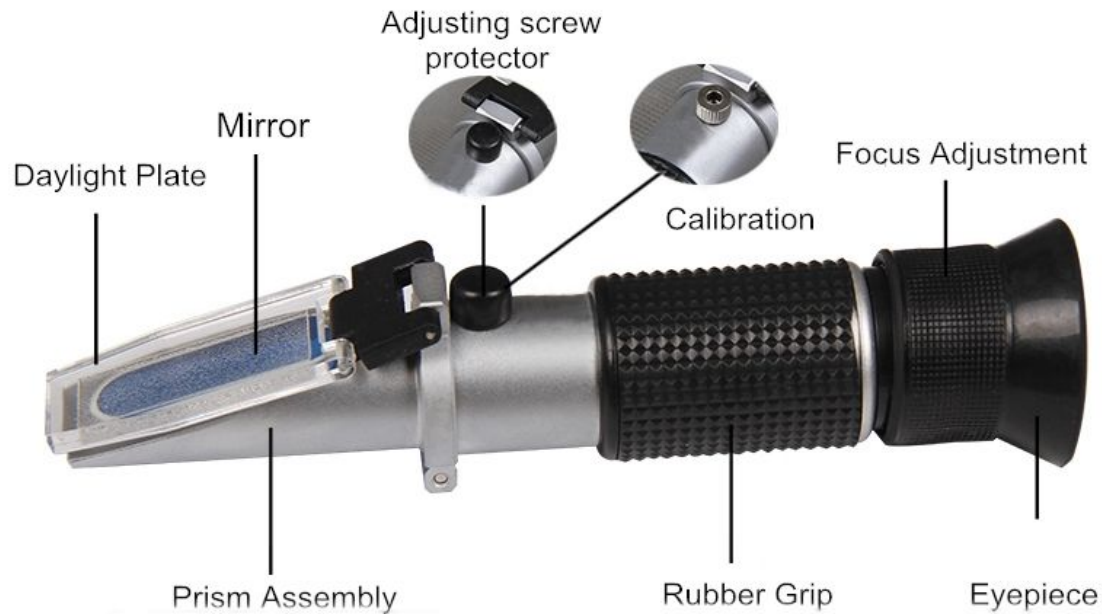
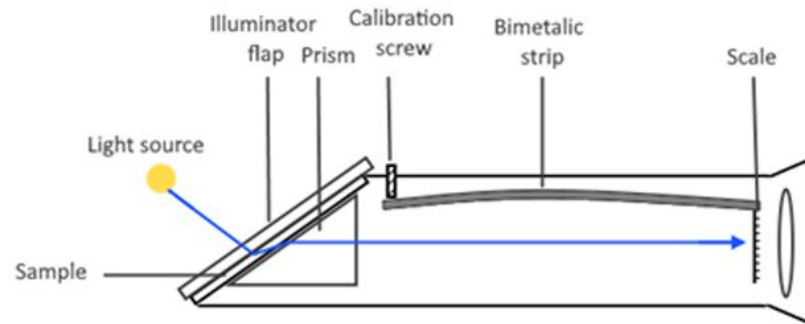
is the change in the direction of light as it speeds up or slows down when moving from one medium to another.

Refractive index is a technique used to study how light bends as it passes from one substance to another.

Any substance through which light can pass has its own refractive index.

Immersion method: Glass particles are placed in a liquid whose refractive index is adjusted until it equals that of the glass. At this point, *the Becke line* (a bright halo at the particle's edge) disappears, indicating equal refractive indices.





Method: The refractive index of the liquid is adjusted by changing temperature. A hot stage apparatus heats the liquid (usually silicone oil) at 0.2°C per minute. The glass refractive index remains stable, while the liquid's index decreases ~ 0.0004 per $^{\circ}\text{C}$. Observations are made under a microscope with sodium D light or other wavelengths.

Result : If fragments show similar match points, they have comparable refractive indices. Exact refractive index values can be calculated at the match point temperature. The refractive index of glass may vary upto 0.0002. For reliable comparison, differences must exceed this threshold to confirm different origins.

Becke line

If the refractive index (n) of glass is different from that (n) of water/liquid medium,

A halo-like ring appears around the edge of the glass. This halo effect is called a Becke line.

It appears because the refracted light becomes concentrated around the edges of the glass fragments.



Becke line	Located inside glass perimeter	n of glass is $>$ surrounding liquid
	Located outside glass perimeter	n of glass is $<$ surrounding liquid
	Glass disappears	Same refractive index of glass & liquid

GRIM

GRIM (*Glass Refractive Index Measurement*) is a computer/video system.

Automates refractive index determination of glass fragments using the immersion method with a hot stage.

Method: A video camera monitors glass fragments as immersion oil is heated or cooled. The system measures image contrast continuously. When contrast reaches a minimum (match point), the refractive index of the glass equals that of the liquid.

The match point temperature is converted into an exact refractive index using stored calibration data.

Birefringence

Normal glass (like soda-lime window glass) is **isotropic**: it has the same refractive index in all directions, so light passes through uniformly.

Birefringence is the division of a ray of light into two rays when it passes through an optically anisotropic material, depending on the polarization of the light. Two different refractive indices are assigned to the material for different polarizations.

Instrument: polarized light microscope

Forensic application:

- Identifying tempered vs. non-tempered glass by stress patterns.
- Revealing fracture stress directions in broken glass.
- Comparing glass fragments by their birefringent properties to link suspects to crime scenes

Physical measurements in glass analysis

1. **Thickness** : Measured using calipers or micrometers. Helps distinguish between types of glass (e.g., window vs. bottle glass). Must be measured on parallel sides for accuracy.
2. **Type of glass** : classified as flat, leaded, tempered, laminated, etc. Each type has distinct fracture patterns and physical properties of its own.
3. **Surface features** : Includes scratches, coatings, weathering, or manufacturing marks. Examined under stereomicroscopes or polarized light microscopes. Can help determine the glass's use (e.g., automotive, architectural).
4. **Colors**: Observed visually or under magnification. Can indicate the presence of certain elements or coatings. UV light may reveal fluorescence or hidden treatments.

Colors

- It is the visible hue of glass, influenced by impurities, that imparts unintentional color.
- Fe → greenish/bluish/gold tint,
- Co → blue,
- Cu/Se → red
- Difference in color = different specimen
- Assessment—visually against a white background in natural light; side-by-side comparison.

Fluorescence

- The glass glows under UV light due to certain chemical additives.
- Atoms & molecules absorb high-energy light, typically UV (254 nm), and then rapidly emit lower-energy visible light.
- Certain chemical components or surface treatments cause the material to glow under UV exposure.

Eg. Mn, Cerium, Co, Cu/Se, Fe; phosphors



Luminescent - Glass bead set



Detection of curvature

The radius of curvature of glass fragments with curved surfaces is measured using a **spherometer**.

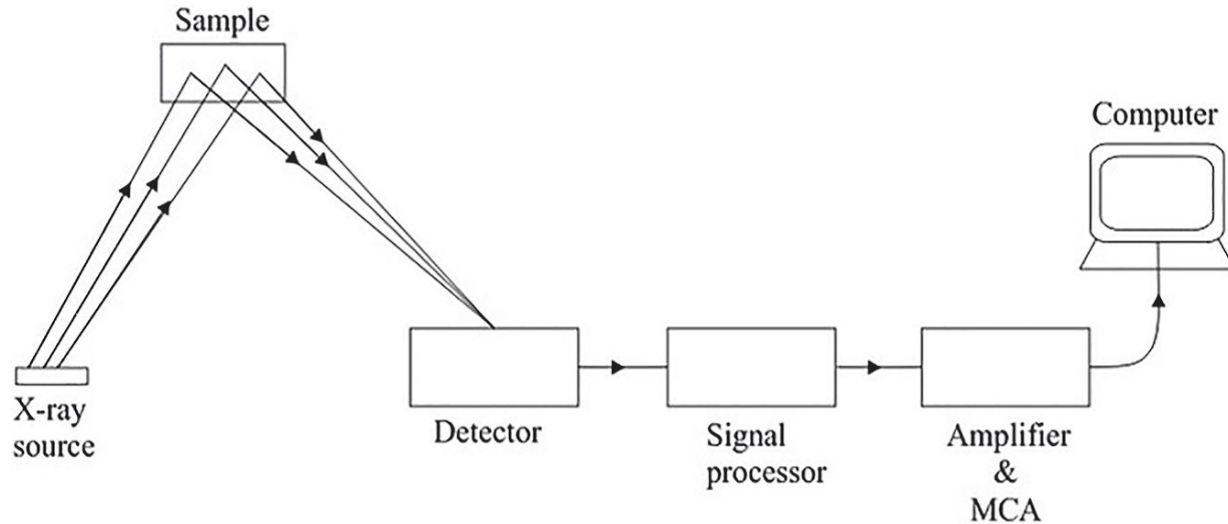
Evidence showing curved surfaces:
Containers, windshield, non-flat surfaces,
decorative items



Elemental analysis

1. **XRF (X-ray Fluorescence Spectrometry):** It is a non-destructive method and measures secondary X-rays emitted by atoms when excited by a primary X-ray source. XRF is good for bulk analysis. Detects elements like Si, Ca, Na, K, Fe.

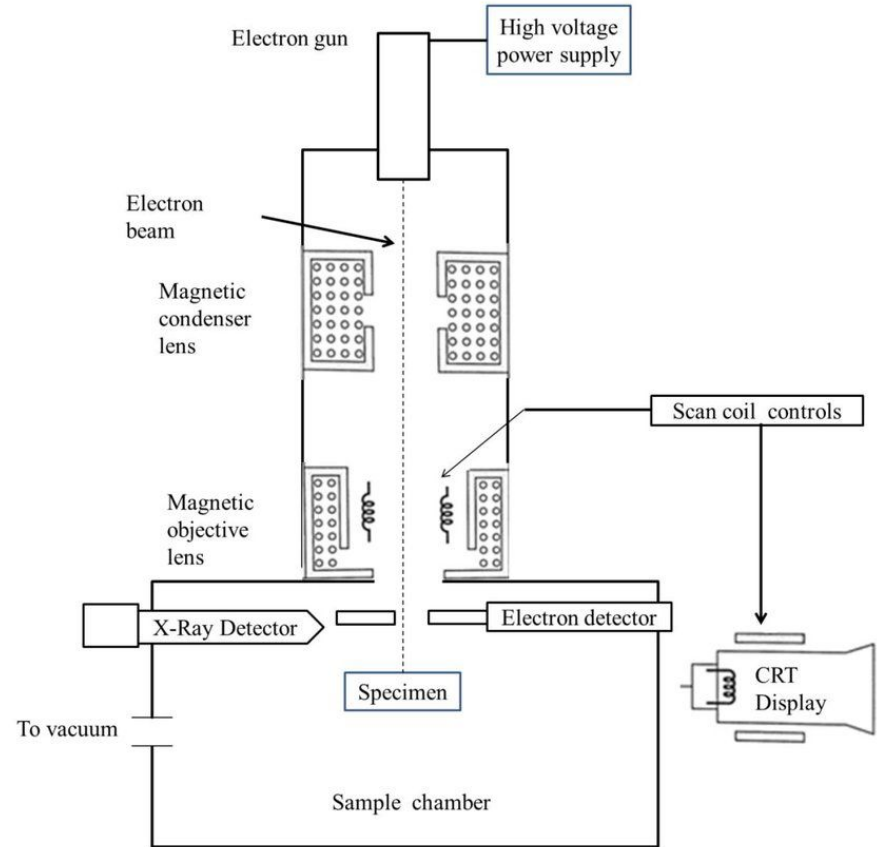
sensitivity = moderate



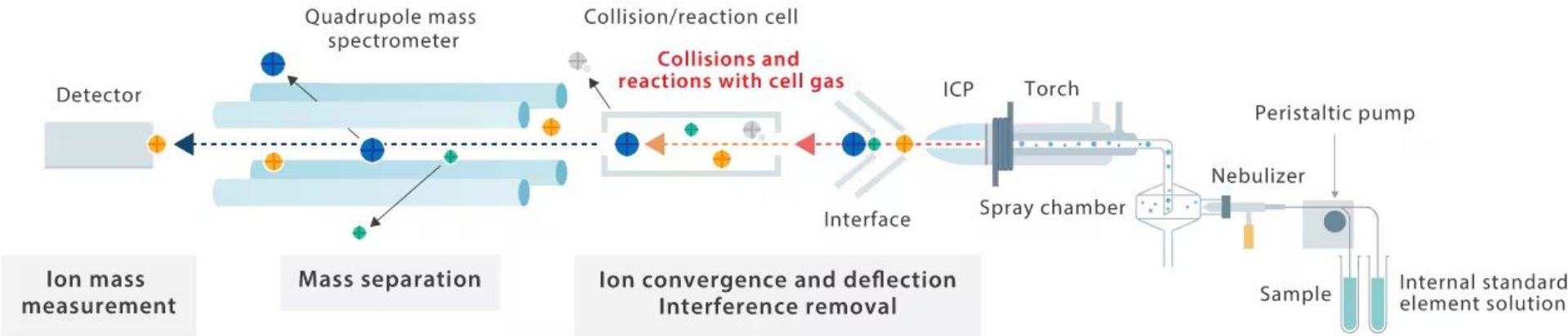
2. SEM-EDS (Scanning Electron Microscopy – Energy Dispersive X-ray Spectrometry): Detects elemental composition by measuring X-rays emitted when the sample is bombarded with electrons.

It provides elemental composition at the micro level and can analyze tiny fragments (<1 mm).

Often paired with high-resolution imaging of surface features.



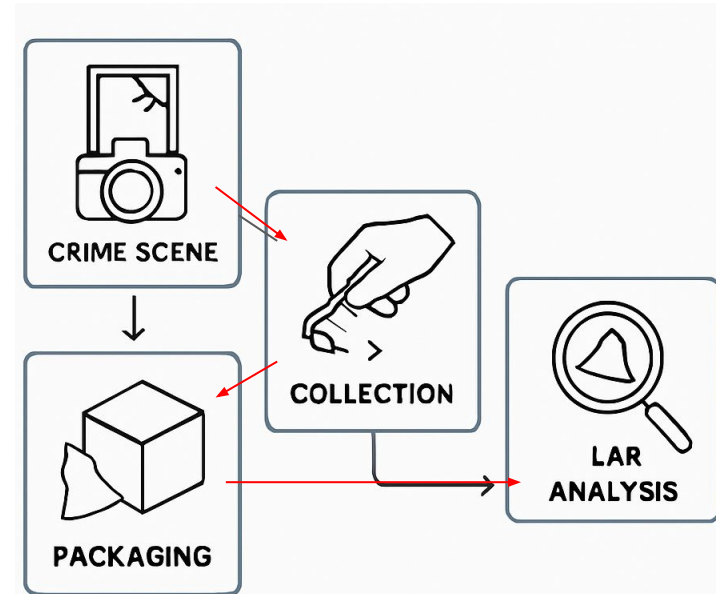
3. ICP-MS (Inductively Coupled Plasma – Mass Spectrometry): The sample is dissolved and introduced into plasma; ions are separated and measured by mass spectrometer. It is extremely sensitive and capable of detecting trace elements at very low concentrations. Often used when high precision is required. **Destructive**



Glass as evidence: by analyzing the fracture patterns, forensic investigators can determine what object hit the glass, from which direction the object came, the sequence of events, and the composition of specialized glass.

Handling of glass samples at crime scenes:

1. Identify and photograph
2. Collect the largest fragments first.
3. Identify the outside and inside surface of any glass fragment
4. If multiple panes are involved make a diagram
5. Note trace evidence
6. Package all materials to maintain the chain of custody.



When broken glass is found at a crime scene, it is gathered and checked for several things:

- Puzzle pieces fitting together.
- Fracture pattern.
- Blood or any DNA source
- Fingerprints
- Composition or type of glass
- Density – mass per unit of volume.
- Determining refractive index , using refractometer
- Any unique characteristics: ex., paint, scratches

Collection of Glass evidence from the crime scene

- Standard reference glass should be taken from the crime scene
- Package in solid containers to prevent breakage
- Preserve garment (shoe, pants, shirt) with glass on it
- All broken glass must be recovered and submitted for analysis when direction of impact is desired.
- Whenever possible, the exterior and interior surfaces of the glass must be indicated. The presence of dirt, paint, grease or putty may indicate the exterior surface of the glass.

Case study format

1. Title
2. Background / Introduction
3. Case Scenario
4. Evidence Description
5. Methods of Examination
6. Results
7. Interpretation
8. conclusion/summary
9. References